

RCF v2.0 Reviews → v2.1 Patch Plan

Consolidation of issues raised by 6 external reviewer voices on RCF v2.0, deduplicated and classified by severity (5 fatal, 8 material, 2 cosmetic). Each issue includes a specific fix proposal naming the exact section, equation, or table to change. Cross-referenced against the v1.0 → v2.0 change log. Includes 8 validations to preserve and a minimum-viable-v2.1 execution checklist.

15
OPEN ISSUES

5
FATAL

8
VALIDATIONS

9
MANDATORY
PATCHES

AUDIENCE
PATCH DEPTH
DATE

Internal Framework Team
Specific · Section / Equation / Table
2026-08-02

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Chapter 1 — Executive Summary

This document consolidates the issues raised by 6 distinct external reviewer voices on RCF v2.0, deduplicates them, classifies them by severity, and proposes a specific fix for each. The reviewers collectively agree that v2.0 is **architecturally stronger than v1.0** — the strip-and-defer acyclicity, the M_{red} definition, the FIREWALL guardrail, the dark energy mechanism, and the self-auditing machinery are all genuine advances. But the reviewers also converge on a clustered set of **status-discipline issues** where v2.0's self-descriptions are not fully earned.

The headline numbers: **15 open issues** (5 fatal, 8 material, 2 cosmetic), plus 8 validations to preserve in v2.1. Of the 15 issues, the v1.0→v2.0 change log I built earlier flagged 2, partially flagged 4, and missed 9. The most significant missed catches are **F2** (K_{ω} is not genuinely built from the Jordan product, despite Remark 3.1.3's claim) and **F4** (T-4 is labeled CLOSED but Z-invariance depends on T-2). Both are cases where v2.0 makes a structural claim and the change log reported the claim without verifying it.

The Single Biggest Risk

T-2 (the stable-mode assumption / infinite-dimensional spectral gap of R_t on $\ker(M)$) is a single point of failure for the majority of the framework's physical predictions: the Convergence Theorem, thin=full equivalence, mass-burden identity, Dirac bracket, $\Lambda_B = 0$, Einstein closure, Friedmann equation, metric boundedness, and dimensional suppression to 2D all inherit T-2. v2.0 is honest about this in §7.2, but a reader who only reads the abstract or section introductions would not know that all GR-level predictions are T-2-conditional. Fix F1 (prominent T-2 callout in abstract + intros + inline tags) is the highest-priority patch.

The reviewers' convergent verdict: v2.0 is a real architectural correction, not a polish pass. The remaining work is mostly **status discipline** — do not call things closed if they still depend on T-2, do not call calibrated constants derived, and keep the dark-energy observational map conditional until the FLRW and Γ_{eff} computations are actually done. Of the 15 open issues, 9 are quick wins (1-line edits, table re-tags, find-and-replace), 3 are medium-effort (paragraph rewrites, quarantine re-tagging), and 3 are hard (new derivations or deep structural work). The minimum viable v2.1 (9 patches) can be assembled from the quick wins and medium-effort honesty fixes; the hard patches (F2 K_{ω} Jordan-built, F3 RP derivation, M5 Friedmann execution) can wait for v2.2 / Phase C.

Chapter 2 — Methodology

The issues were consolidated by: (a) reading all reviewer voices in the uploaded file and the accompanying message body; (b) deduplicating issues raised by multiple reviewers; (c) classifying by severity using a 3-tier rubric; (d) for each issue, naming the exact section, equation, or table to change and the exact old → new text where applicable; (e) cross-referencing each issue against the v1.0→v2.0 change log to note whether it was previously flagged.

2.1 Reviewer Voices

Six distinct reviewer voices were identified in the uploaded material. They are merged into one consolidated voice in this document, with attributions noted per issue (R1, R2, etc.).

ID	Voice	Contribution
R1	Comprehensive review (section-by-section, in uploaded file)	Section-by-section analysis of all 7 v2.0 sections; thematic breakthroughs; open problems; comparison to LQG/CST/CDT/AS/QFT; Phase C recommendations
R2	Executive verdict (in message body)	5 main issues: mass-burden inconsistency, T-4 tension, dark-energy status, quarantine convention, $\Pi_{\max}/\kappa_{B/C}=8\pi$
R3	Formal architectural review (12-section, in message body)	12-section formal assessment; critical concerns (T-2 bottleneck, RP status, D=3 three-channel, §2–§3 interdependence); Phase C readiness checklist
R4	K _ω critic (in message body)	K _ω not genuinely Jordan-built (Def 3.1.2 + Remark 3.1.3); A _{loc} selection language fixed (noted as progress)
R5	Final verdict (in message body)	Validations to preserve; quarantine convention (OPEN/STRUCTURAL/CONDITIONAL); Phase C next steps (T-2 first, dark energy letter, Friedmann, notation pass, S _{eff} rigor)
R6	OneDrive-linked reviewer (in message body)	Section-by-section table with brief reviews; 5 main issues; 'best structural improvement' and 'most important next work'

2.2 Severity Rubric

Issues are classified into three severity tiers based on the consequence of NOT fixing them:

Severity	Definition	Color
FATAL	Would falsify the framework's central claims or break internal coherence if not fixed. Examples: a structural claim that is named but not derived (F2, F3); a status label that contradicts the dependency graph (F4); a single point of failure not surfaced to readers (F1); an ad hoc assumption at the foundation (F5).	

Severity	Definition	Color
MATERIAL	Would mislead readers about what is established vs conditional. Examples: inconsistency across sections (M1); status inflation by inheritance (M2); convention ambiguity (M3); 'derived' vs 'calibrated' language (M4); derivation not executed (M5); interdependence tighter than claimed (M6); generous status label (M7); dependency not propagated (M8).	
COSMETIC	Typos, stale references, naming issues, LaTeX artifacts. Do not affect correctness but must be cleaned before external dissemination. Examples: LaTeX artifacts (C1), misleading theorem name (C2).	

2.3 Cross-Reference Protocol

Each issue is cross-referenced against the v1.0→v2.0 change log I built earlier (RCF_v1_to_v2_Change_Log.pdf, 48 pages). The cross-reference uses three categories: **Flagged** (the change log identified the issue and described it accurately), **Partially flagged** (the change log noted the relevant fact but did not connect it to the issue), **Not flagged** (the change log missed the issue entirely). The summary is in Chapter 7.

Chapter 3 — Validations to Preserve in v2.1

The reviewers praised 8 things about v2.0 that the v2.1 patch plan must NOT weaken. These validations are listed below with: what they are, why reviewers praised them, and what v2.1 must not touch. The patches in Chapters 4–6 are designed to add tags, downgrade overclaims, and surface dependencies — not to remove the structural advances that earned the validations.

3.1 V1 — The Self-Auditing Mechanism (§7)

Location: §7 (entire section) | *Reviewers:* R1, R3, R5

What it is. The six-category status taxonomy (Def 7.1.1), the consolidated theorem table (Table 7.1.2), the seven theorem targets T-1..T-7, the 17-point quarantine list, the formal action S_{eff} , the acyclic dependency graph verification (Thm 7.5.1), and the five final guardrails.

Why praised. Multiple reviewers called this 'exemplary' and 'a standard for theoretical manuscript self-policing.' The discipline of tagging every claim with an explicit epistemic status, cataloguing stale cross-references, and maintaining a quarantine list prevents overclaiming and makes the framework's epistemic posture checkable.

Preserve. Retain verbatim as a template. Do not weaken the status taxonomy, do not collapse the quarantine list, do not retire the guardrails. The v2.1 patch plan adds tags and downgrades, but does not remove the audit structure that makes those tags meaningful.

3.2 V2 — The FIREWALL Guardrail (Princ 2.8.1)

Location: §2.8.1 | *Reviewers:* R1, R3, R5

What it is. The strict separation between probabilistic branch weights p_k (Layer A) and algebraic burden B_{Δ} (Layers A/B/C/Q'). Burden linearity is a proven algebraic identity (Tr is linear, F is fixed), not a probabilistic average. Cross-sector gravity is quarantined because it would require either prohibited probabilistic averaging or a new algebraic coupling not currently present.

Why praised. Called 'conceptually vital and well-executed' and 'one of the framework's most important architectural protections.' It prevents the smuggling of quantum probability into gravitational sourcing — a subtle error that has undermined other emergent gravity programs.

Preserve. Do not soften the FIREWALL. The v2.1 patch plan's T-4 downgrade (F4) affects the Born rule's epistemic status, not the FIREWALL separation. The FIREWALL holds regardless of whether T-4 is CLOSED or STRENGTHENED: burden linearity is an algebraic identity independent of the Born rule's derivation path.

3.3 V3 — The Dark Energy Mechanism (§2.2 + §6.3)

Location: §2.2 (Defs 2.2.1, Prop 2.2.2, Conj 2.2.3) + §6.3 (Conj 6.3.2, Thms 6.3.3–6.3.5, Cor 6.3.6) | *Reviewers:* R1, R3, R5

What it is. The structural form $\rho_{\text{DE}} \sim (H/\Gamma_{\text{eff}})^2 \cdot \rho_{\text{source}}$ with $\Gamma_{\text{eff}} \sim \eta \cdot B_{\Delta}$. The 120-order resolution as $(H_0 t_P)^2 \sim 10^{-122}$. The Γ evolution argument ($\rho_{\text{DE}} \sim (1+z)^{-3}$, $\Omega_{\text{DE}} \sim (1+z)^{-6}$ at recombination). The equation of state $w > -1$ always, evolving from 0 to -1 .

Why praised. Called 'the framework's most distinctive and testable prediction' and 'the strongest physical prediction of the framework.' The 120-order suppression falls out structurally (exponent 2 from $B_{\Delta} = \|\text{mismatch}\|^2$), not by tuning. The $(1+z)^{-6}$ EDE suppression resolves the CMB $\Omega_{\text{EDE}} < 0.004$ bound without fine-tuning.

Preserve. Do not weaken the structural mechanism. The v2.1 patch plan's M2 fix (dark-energy cosmological predictions status) only adds conditional tags to the cosmological APPLICATION — the structural scaling laws in §2.2 remain Established. Ensure the mechanism is prominently featured in the Phase C abstract.

3.4 V4 — The Epistemic Taxonomy (Def 7.1.1)

Location: §7.1, Definition 7.1.1 | *Reviewers:* R3, R5

What it is. Six categories: Established (proven from primitives with no open hypotheses), Conditional (proven modulo explicitly stated assumptions), Conditional→Strengthened (strengthened by derivation), Theorem Target (open proof obligation with a T-ID), Structural (definition or framework convention), Conjecture (quarantined claim with a Q-ID). CLOSED is reserved for results with zero open blocking dependencies.

Why praised. Called 'excellent' and 'a model of disciplined exposition.' The taxonomy makes it impossible to accidentally overclaim — every result must wear its epistemic status on its sleeve.

Preserve. Retain the six-category taxonomy. The v2.1 patch plan's M3 fix (quarantine convention) adds a three-hard-status sub-convention for the quarantine list (OPEN / STRUCTURAL / CONDITIONAL) but does not replace the top-level taxonomy.

3.5 V5 — The M_{red} Definition (Def 1.4.3)

Location: §1.4.3 | *Reviewers:* R1, R5

What it is. $M_{\text{red}} = \sum_{\{i \in I_2\}} w_i \cdot D_i$ if $D_i \in A_{\text{red}}$, framed as 'the SEPARATION of M into its Tier 1 and Tier 2 sub-collectives,' not a 'reduction.' The full $M = M_{\{\text{Tier1}\}} + M_{\text{red}}$; on A_{red} , only M_{red} survives.

Why praised. Called 'the single largest v1.0 definitional gap, now closed.' The downstream effect: the sectorwise zero-decomposition (Thm 2.7.1) now has genuine content (the p_k are constrained by M_{red} 's sectoral structure, not arbitrary), which gives the Born rule derivation a real objectivity premise.

Preserve. Do not redefine M_{red} . The 'separation, not reduction' framing is load-bearing — it pre-empts a reader misreading of M_{red} as a 'reduced' version of M . The v2.1 patch plan's F4 fix (T-4 downgrade) does not touch M_{red} 's definition; it only adds a conditional tag to the downstream Born rule.

3.6 V6 — The Lie–Jordan Decomposition (Def 0.1.2)

Location: §0.1.2 | *Reviewers:* R1, R5

What it is. $AB = \frac{1}{2}[A, B] + (A \circ B)$, equipping A with two genuinely independent algebraic structures: a Lie algebra A_L (antisymmetric, ordering) and a Jordan algebra A_J (symmetric, co-occurrence). Jordan–von Neumann–Wigner (1934) asserts their mathematical independence.

Why praised. Called 'the single most important structural improvement' and 'a rigorous algebraic foundation for the two primitives.' It provides a clear answer to 'why are causality and locality separate?' — because the Lie and Jordan structures of the algebra are separate.

Preserve. Do not modify the decomposition. The v2.1 patch plan's F2 fix (K_ω not Jordan-built) addresses whether K_ω actually uses the Jordan product in its formula — it does NOT challenge the Lie–Jordan decomposition itself. The decomposition stays; the claim about what K_ω inherits from it must be made honest.

3.7 V7 — The Acyclicity Verification (Thm 7.5.1)

Location: §7.5, Theorem 7.5.1 + Table 7.5.2 | *Reviewers:* R1

What it is. The cumulative dependency graph across Sections 0–6 is verified acyclic. v1.0 had only asserted acyclicity via narration; v2.0 promotes it to a formally stated theorem with an itemized verification and a structured table enumerating 11 forward-reference resolutions.

Why praised. Called 'a meaningful improvement over v1.0's narrated assertion.' The verification is checkable: each section's dependencies are itemized by section number, and each forward reference has a status (RESOLVED / CERTIFIED / PARTIALLY RESOLVED).

Preserve. Do not weaken the verification. The v2.1 patch plan's M6 fix (§2–§3 interdependence) acknowledges that the one-way reference structure is tighter than a pure dependency chain suggests, but this does not break acyclicity — it refines the commentary around it.

3.8 V8 — The SOE/MOE → Open Expansion/Open Extension Rename

Location: §1.5.2 (Open Expansion Flow), §1.5.3 (Open Extension Flow) | *Reviewers:* R1, R6

What it is. v1.0's 'Single Open Extension' (SOE) is renamed 'Open Expansion' — the local, isometric expansion of locality-built sectors. v1.0's 'Multiple Open Extension' (MOE) is renamed 'Open Extension' — the global, contractive extension of reconciliation across accumulated expansions. The rename captures the actual picture: sub-sectors do not have separate causal layers; they all share the single causal layer and expand to reconcile with it.

Why praised. Called 'more than cosmetic — it encodes the corrected picture.' The mathematics is unchanged; the physical interpretation is sharpened. Definition headers carry '(was SOE)' / '(was MOE)' parentheses for traceability.

Preserve. Do not revert the rename. The v2.1 patch plan does not touch terminology. The rename propagates consistently through every section (Arrow of Time generator assignment, Friedmann as 'Open Extension descent on FLRW,' etc.) and must remain consistent.

Chapter 4 — Fatal Issues (5)

Five issues would falsify the framework's central claims or break internal coherence if not fixed. Each is presented with: the exact v2.0 location, the problem (what is wrong), the specific fix proposal (old → new text where applicable), the fix difficulty, whether it blocks Phase C, and a cross-reference to whether the v1.0→v2.0 change log flagged it.

Fatal Issue Summary

F1 T-2 as single point of failure — not surfaced in abstract/intro. **F2** K_ω is not genuinely built from the Jordan product (Def 3.1.2 + Remark 3.1.3). **F3** Reconciliation Principle not derived from Open Expansion (violates Guardrail 4). **F4** T-4 overclaim — CLOSED but Z-envariance depends on T-2. **F5** D=3 three-channel assumption is ad hoc (not derived from algebra).

► [F1] T-2 as single point of failure — not surfaced in abstract/intro

Severity: FATAL | Reviewers: R1, R3, R5 | v2.0 location: Abstract and introductions of §§1–7 (the dependency is documented in §7.2 but buried)

Problem. T-2 (the stable-mode assumption / infinite-dimensional spectral gap of R_t on $\ker(M)$) underpins: the Convergence Theorem (Thm 1.6.1), the thin=full equivalence (Thm 1.7.4), the mass-burden identity (Thm 2.5.1), the emergent Dirac bracket (Thm 1.4.6), $\Lambda_B = 0$ (Cor 4.5.5), the Einstein closure (Thm 4.5.4), the Friedmann equation (Conj 6.2.2), metric boundedness / singularity avoidance (Thm 4.7.1, Thm 5.2.1), and the dimensional suppression to 2D (Thm 5.3.1). If T-2 fails, the majority of the framework's physical predictions collapse from Established to Conditional. The framework is honest about this in §7.2, but a reader who only reads the abstract or section introductions would not know that all GR-level predictions are T-2-conditional.

Fix. Add a prominent T-2 dependency callout to: (a) the manuscript abstract (single sentence: 'All GR-level predictions (§4–§6) are conditional on T-2, the stable-mode assumption; see §7.2.');

(b) the introduction of §4 (Time, Gravity), §5 (Black Holes), and §6 (Cosmology) — each should open with a one-sentence T-2 flag; (c) every theorem whose proof invokes T-2 — add an inline [T-2] tag next to the theorem statement, not just in the status column of Table 7.1.2.

Fix difficulty: Medium (paragraph-level edits across 4 sections + abstract) | Blocks Phase C: Yes — Phase C cannot claim 'mathematically established' without T-2 closure or prominent T-2 flagging

Change-log cross-ref: Partially flagged

The v1.0→v2.0 change log (Ch 6) listed T-2 as the existential open target and noted its blocking role. But the change log did NOT flag that the abstract and section introductions fail to surface this dependency — a reader-facing issue, not a structural one.

► [F2] K_ω is not genuinely built from the Jordan product

Severity: FATAL | Reviewers: R4 | v2.0 location: Def 3.1.2 (K_ω formula) + Remark 3.1.3 (Jordan-structure claim)

Problem. Def 3.1.2 gives $K_\omega(A, B) = |\omega_{\text{kin}}(A \dagger B)|^2 / [\omega_{\text{kin}}(A \dagger A) \cdot \omega_{\text{kin}}(B \dagger B)]$ — the raw sesquilinear form, no $A \circ B$ anywhere. Remark 3.1.3 then claims the symmetry 'comes from the Jordan structure,' but the proof it gives is $|\omega(A \dagger B)| = |\omega(B \dagger A)| = |\omega(A \dagger B)^*| = |\omega(A \dagger B)|$ — pure state-Hermiticity, never touches $A \circ B$. K_ω would be exactly as symmetric with zero reference to the Jordan structure. The label is asserted, not

shown. This is the same pattern as the T-4 and M_{red} gaps from v1.0 — a structural claim that is named but not derived — on the locality side this time.

Fix. Two options, pick one: (a) **PREFERRED** — build a K_ω -analogue with $A \circ B$ in the formula itself, parallel to how $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$ sits directly inside F's formula. Candidate: $K_\omega^J(A,B) = \omega_{\text{kin}}(A \circ B) / \sqrt{(\omega_{\text{kin}}(A^\dagger A) \cdot \omega_{\text{kin}}(B^\dagger B))}$, which genuinely uses the Jordan product and is symmetric by construction. Then show K_ω^J reduces to the current K_ω under the GNS state. (b) **MINIMAL** — honestly downgrade Remark 3.1.3 to: 'K_ω's symmetry comes from state-Hermiticity ($|\omega(A^\dagger B)| = |\omega(B^\dagger A)|$). The Jordan structure licenses the interpretation of K_ω as a correlation (symmetric, undirected) but does not generate the symmetry.' Option (a) is the right fix; option (b) is the honest fallback if (a) proves technically hard.

Fix difficulty: Hard (requires constructing a new kernel and proving reduction) | **Blocks Phase C:** Yes — the locality primitive's algebraic grounding is incomplete until K_ω genuinely uses $A \circ B$ or the claim is retracted

Change-log cross-ref: Not flagged

The v1.0→v2.0 change log (Ch 4.4, Change 10) noted that v2.0 §3 added Remark 3.1.3 'ties K_ω to the Jordan structure' — but took v2.0's self-description at face value. The change log did not verify that the tie is genuine (i.e., that $A \circ B$ actually appears in K_ω 's formula). This is the most significant missed catch.

► [F3] Reconciliation Principle not derived from Open Expansion

Severity: FATAL | *Reviewers:* R3 | **v2.0 location:** Princ 2.1.1 (RP statement) + Post 1.1.1 (Open Expansion Principle) + Guardrail 4 (§7.6.1)

Problem. The Reconciliation Principle (Princ 2.1.1) is labeled 'Structural,' not 'Established.' Its relationship to the Open Expansion Principle (Post 1.1.1) is not derived. The manuscript frames RP as 'the dynamic consequence' of Open Expansion, but the derivation path is not explicit. If RP is irreducible, it is a new primitive — violating Guardrail 4 ('No new primitives without explicit justification'). If RP is derived, the derivation is missing. Either way, the framework has more primitives than its own guardrails permit.

Fix. Two options: (a) **DERIVE** RP from Open Expansion — show that the variational target $I(S) = \sum [s(a,b) - s^*(a,b)]^2$ is the unique functional whose minimization is equivalent to enforcing Lie–Jordan compatibility under finite-rate Open Expansion. This would make RP a theorem, not a primitive. (b) **EXPLICITLY LIST** RP as a primitive alongside Postulate 1.1.1 in §7.6.1 Guardrail 4, and update the guardrail to read: 'The framework's primitives are: A, the Lie–Jordan decomposition, admissibility, M, F, causality, locality, the Open Expansion Principle, and the Reconciliation Principle.' Option (a) is preferred; option (b) is the honest fallback.

Fix difficulty: Hard (requires a derivation) or Easy (add RP to the primitive list) | **Blocks Phase C:** Yes — the framework's primitive count must be honest before Phase C can claim minimality

Change-log cross-ref: Not flagged

The v1.0→v2.0 change log (Ch 4.3) noted that RP was moved to §2.1 and that v2.0 frames it as 'the dynamic consequence' of Open Expansion, but did not flag that the derivation is missing. The change log was descriptive (reported v2.0's framing) not evaluative (did not check whether the framing is earned).

► [F4] T-4 overclaim — CLOSED but Z-envariance depends on T-2

Severity: FATAL | *Reviewers:* R2, R3, R5 | **v2.0 location:** Table 7.2.1 (T-4 row: 'CLOSED') + Def 2.7.3 (Z-envariance: 'Theorem Target T-2')

Problem. §7.2.3 says T-4 is CLOSED and the Born rule is now a theorem of RCF, not an axiom. But §2.7.3 says Z-envariance is a Theorem Target T-2 because the full proof requires showing that label swaps commute with the Open Extension vector field. If Z-envariance depends on T-2, and the Born rule (Thm 2.7.4) depends on Z-envariance, then T-4 is closed modulo T-2, not unconditionally closed. The '(CLOSED)' label in Table 7.2.1 is too strong. This is the same category of self-contradiction as v1.0's 'CLOSED (conditional on T-2)' for Q-9 — which v2.0 itself corrected.

Fix. Change Table 7.2.1 row T-4 from 'CLOSED' to 'STRENGTHENED (conditional on T-2)'. Add a remark to §7.2.3: 'T-4 is strengthened relative to v1.0 (the objectivity premise now has genuine content via M_red), but remains conditional on T-2 because Z-envariance (Def 2.7.3) requires the Open Extension vector field to commute with label swaps, which is T-2-conditional.' Alternatively, explicitly separate 'Born-rule functional derivation' (closed: given Z-envariance, $p_i = |c_i|^2$ follows) from 'Z-envariance derivation' (conditional on T-2: the label-swap symmetry of the Open Extension flow).

Fix difficulty: Quick win (1-line table edit + 1-paragraph remark) | **Blocks Phase C:** No — but must be fixed before Phase C can claim honest status labels

Change-log cross-ref: Not flagged

The v1.0→v2.0 change log (Ch 4.8, Change 4) noted that v2.0 says 'T-4 is CLOSED' and that Z-envariance depends on T-2, but did not flag the contradiction between these two statements. The change log reported both facts without connecting them.

► [F5] D=3 three-channel assumption is ad hoc

Severity: FATAL | *Reviewers:* R1, R3 | *v2.0 location:* Def 3.6.1 (three inference channels: existence, position, direction) + Thm 3.6.4 (D=3 closure) + Remark 3.6.5

Problem. Theorem 3.6.4 (D=3 closure, demoted to Conditional Proposition) requires that existence, position, and direction are the only inference channels. This assumption is not derived from the algebra; it is imposed structurally. The manuscript acknowledges this in Remark 3.6.5 ('A skeptic could ask: why can't there be a 4th inference channel?'), but the answer ('structural') is not a proof. For a framework whose central claim is the emergence of 3D space from algebraic primitives, this is a significant outstanding gap. The D=3 demotion from Theorem to Conditional Proposition fixed the tautological $\Xi_C(D) = |D-3|$ problem, but did not fix the deeper three-channel assumption.

Fix. Expand Remark 3.6.5 into a full discussion (target: 1 page) covering: (a) why the three-channel assumption is not derivable from the current algebraic primitives; (b) what would falsify it (e.g., a fourth inference channel emerging from a higher-order correlation structure); (c) what would derive it (e.g., a rank argument on the cubic Gram matrix that forces $\text{rank} \leq 3$ from algebraic constraints, not just from observing that three channels exist); (d) whether the proposition should be further demoted from 'Conditional Proposition' to 'Structural Conjecture' pending such a derivation. If (c) cannot be completed in Phase C, explicitly flag D=3 as the framework's second-largest open problem after T-2.

Fix difficulty: Medium (1-page discussion expansion) or Hard (if attempting the rank derivation) | **Blocks Phase C:** No — but the honest discussion must be present before Phase C can claim the framework 'derives' 3D space

Change-log cross-ref: Partially flagged

The v1.0→v2.0 change log (Ch 4.4, Change 1) noted the D=3 demotion and the tautological Ξ_C fix, but did not flag that the three-channel assumption itself is ad hoc. The change log focused on the Ξ_C definition fix (which v2.0 explicitly self-documents) and missed the deeper assumption that Ξ_C 's rank argument depends on.

Chapter 5 — Material Issues (8)

Eight issues would mislead readers about what is established vs conditional. Same structure as Chapter 4.

Material Issue Summary

M1 Mass-burden identity inconsistency (§1 says $m \equiv B_0$, §2 says $m^2 \equiv B_0$). **M2** Dark-energy cosmological predictions status too strong (§6 labels 120-order/EDE/ $w(z)$ as Theorem while Friedmann is Conjecture). **M3** Quarantine arithmetic convention not finalized (PARTIAL vs OPEN boundary fuzzy). **M4** $\Pi_{\max}/\kappa_B/C=8\pi$ presented as 'derived' in some summaries (should be 'calibrated'). **M5** Friedmann derivation not executed (Conj 6.2.2 path stated but not performed). **M6** §2–§3 interdependence tighter than acyclicity suggests. **M7** T-1 status generous (Thm 1.1.2 is dimensional argument, not derivation). **M8** Open Extension Lindblad form T-2 dependency not propagated to downstream theorems.

► [M1] Mass-burden identity inconsistency across sections

Severity: MATERIAL | **Reviewers:** R2 | **v2.0 location:** §1 (forward-reference to §2: says ' $m \equiv B_0$ ') vs §2.5.1 (Thm 2.5.1: says ' $m^2 \equiv B_0$ ') + summary tables in §1 and §2

Problem. §2 says the rewrite fixes the mass-burden identity by stating it consistently as $m^2 \equiv B_0$, and its summary table lists 'Mass-burden identity $m^2 \equiv B_0$.' But §1 still says Section 2 will introduce 'Mass-burden identity — $m \equiv B_0$.' The v1.0 algebraic inconsistency ($m \equiv B_0$ in the headline vs $B_0 = m^2$ in Eq 4.2.2) was fixed in §2.5.1 but not propagated to §1's forward-reference.

Fix. Update §1's forward-reference from ' $m \equiv B_0$ ' to ' $m^2 \equiv B_0$ '. Audit ALL summary tables and cross-references across all 8 sections for the same convention. Search for: ' $m \equiv B$ ', ' $m = B$ ', 'mass-burden identity', ' $m \equiv B_0$ ' — every occurrence should read ' $m^2 \equiv B_0$ ' or ' $m = \sqrt{B_0}$ ' (in natural units) or ' $m = (\hbar_{\text{eff}}/c_{\text{eff}}) \cdot \sqrt{B_0}$ ' (dimensionful).

Fix difficulty: Quick win (find-and-replace + manual verification) | **Blocks Phase C:** No — but must be fixed for internal consistency

Change-log cross-ref: Not flagged

The v1.0→v2.0 change log (Ch 4.3, Change 13) noted the $m \equiv B_0 \rightarrow m^2 \equiv B_0$ fix in §2.5.1, but did not flag that §1's forward-reference still says ' $m \equiv B_0$ '. The change log was section-by-section and did not cross-check forward-references against the referenced section's actual statement.

► [M2] Dark-energy cosmological predictions status too strong

Severity: MATERIAL | **Reviewers:** R2 | **v2.0 location:** §6.2 (Conj 6.2.2: Friedmann is 'Structural conjecture') vs §6.3 (Thms 6.3.3–6.3.5: 120-order, EDE suppression, $w(z)$ labeled as 'Theorem')

Problem. §6.2 says the Friedmann equation is a structural conjecture and that its derivation path has not been performed. But §6.3 then labels the 120-order resolution (Thm 6.3.3), early dark-energy suppression (Thm 6.3.4), and dark-energy equation of state (Thm 6.3.5) as 'Theorem' — implying they are established. But these cosmological predictions inherit the Friedmann conjecture's status: if the FLRW reduction of the ADM ansatz hasn't been performed, the cosmological application of the $\rho_{\text{DE}} \sim (H/\Gamma)^2$ scaling is conditional on that reduction, not established.

Fix. Keep the structural mechanism ($\rho_{\text{DE}} \sim (H/\Gamma)^2$ scaling from §2.2) as Established. But mark the cosmological predictions (120-order resolution Thm 6.3.3, $(1+z)^{-6}$ suppression Thm 6.3.4, $w(z)$ evolution Thm 6.3.5) as 'CONDITIONAL on the FLRW reduction and Γ_{eff} model' — i.e., conditional on Conj 6.2.2 being upgraded from Conjecture to Conditional Theorem. Add a remark to §6.3: 'The scaling laws in §6.3 are structurally derived from the §2.2 mechanism. Their cosmological application assumes the FLRW reduction of the ADM ansatz (Conj 6.2.2), which is not yet performed; until that reduction is completed, these results are conditional on the Friedmann derivation path.'

Fix difficulty: Quick win (add conditional tags + 1-paragraph remark) | **Blocks Phase C:** No — but must be fixed before cosmological predictions can be presented as testable without qualification

Change-log cross-ref: Not flagged

The v1.0→v2.0 change log (Ch 4.7) noted the dark energy mechanism and the $(1+z)^{-6}$ suppression as v2.0 additions, but did not flag that §6 labels these as 'Theorem' while the Friedmann equation they depend on is a 'Conjecture.' The change log reported the additions without checking the status inheritance.

► [M3] Quarantine arithmetic convention not finalized

Severity: MATERIAL | *Reviewers:* R2, R5 | *v2.0 location:* §7.3 Remark 7.3.2 (corrected arithmetic: 0 CLOSED, 2 STRUCT, ~11 PARTIAL, ~6 OPEN) + Q-1, Q-8 boundary fuzzy

Problem. §7.3 corrects the v1.0 arithmetic (which claimed '13 PARTIAL' but listed 11 items, summing to $19 \neq 17$) but says the exact boundary between PARTIAL and OPEN is fuzzy for items like Q-1 and Q-8, and that the framework should adopt a consistent convention. The current convention is ambiguous: is Q-1 (dark matter amplitudes) PARTIAL (structural mechanism exists) or OPEN (no quantitative prediction)? Is Q-8 (holography) PARTIAL (2D boundary derived) or OPEN (full AdS/CFT not derived)?

Fix. Adopt three hard statuses only for the quarantine list: (1) OPEN — no quantitative mechanism; (2) STRUCTURAL — mechanism exists, no amplitude/recovery; (3) CONDITIONAL — proof path exists but depends on T-2/T-7/simulation. Reserve CLOSED only for results with no open blockers (as §7.3 already instructs). Apply consistently: Q-1 = STRUCTURAL (mechanism established, amplitudes require simulation); Q-8 = STRUCTURAL (2D boundary derived, full AdS/CFT not); Q-9 = CONDITIONAL (120-order resolved structurally, $\Lambda_B = 0$ conditional on T-2). Update Remark 7.3.2 to use these three statuses explicitly.

Fix difficulty: Medium (re-tag all 17 Q-items against the new convention) | **Blocks Phase C:** No — but must be fixed for quarantine list to be unambiguous

Change-log cross-ref: Partially flagged

The v1.0→v2.0 change log (Ch 4.8, Change 3) noted the arithmetic fix (13→11 PARTIAL), but did not flag that the convention itself is fuzzy. The change log reported the fix without evaluating whether the fixed convention is itself well-defined.

► [M4] Π_{max} , κ_B , and $C=8\pi$ presented as 'derived' in some summaries

Severity: MATERIAL | *Reviewers:* R1, R2 | *v2.0 location:* §5.1.3 ($\Pi_{\text{max}} = C_{\Pi}/\ell_0^2$, C_{Π} open) + §4.5.6 (κ_B , C calibrated by Newtonian) + various summary tables

Problem. §5.1.3 defines $\Pi_{\text{max}} = C_{\Pi}/\ell_0^2$ dimensionally and states the spectral coefficient C_{Π} remains open under T-2. §4.5.6 says $C = 8\pi$ is calibrated by the Newtonian limit rather than derived from saturation. These are honest in-section. But some summary tables and the abstract still use language like ' κ_B derived' or ' Π_{max} resolved' without the 'calibrated' / 'partially resolved' qualifier.

Fix. Audit all summary tables, abstracts, and architectural summaries for ' κ_B derived', ' Π_{max} resolved', ' $C = 8\pi$ derived'. Replace with: ' κ_B : dimensional form derived, $C = 8\pi$ calibrated by Newtonian limit' / ' Π_{max} : dimensional form derived, C_{Π} open (T-2)' / ' $C = 8\pi$: calibrated, not derived'. The in-section language is already correct; the summaries must catch up.

Fix difficulty: Quick win (find-and-replace in summary tables) | **Blocks Phase C:** No — but must be fixed for honest external presentation

Change-log cross-ref: Flagged

The v1.0→v2.0 change log (Ch 4.5, Change 8; Ch 4.6, Change 3) correctly noted that v2.0 presents κ_B as 'calibrated, not derived' and Π_{max} as 'partially resolved, coefficient open.' This is one of the two issues the change log did catch.

► [M5] Friedmann derivation not executed

Severity: MATERIAL | *Reviewers:* R1, R3, R5 | *v2.0 location:* Conj 6.2.2 (4-step derivation path stated, computation not performed)

Problem. §6.2.2 honestly labels the Friedmann equation as a 'Structural conjecture' with the 4-step derivation path stated (FLRW-restricted ADM ansatz → homogeneous/isotropic burden tensor components → S_{eff} variation → $\kappa_B = 8\pi G/c^4$ calibration by Newtonian limit), but admits 'the computation has not been performed.' Two specific gaps: (a) the $8\pi G/3$ coefficient is imported from standard GR, not derived from RCF primitives; (b) the non-local relational channel Θ^{rel} (built from Π_{net} per §3.9) requires FLRW averaging that v1.0 never addressed and v2.0 leaves as part of the open derivation. For Phase C, this derivation must be completed or the cosmological claims remain heuristic.

Fix. For Phase C: either (a) EXECUTE the 4-step FLRW ADM variation to upgrade Conj 6.2.2 to a Conditional Theorem (closes Q-13 to PARTIAL: amplitudes open, structure proven); or (b) MAINTAIN 'Conjecture' status in the final text and do not let it drift toward 'Established' through rhetorical accumulation. Option (a) is preferred; option (b) is the honest fallback. If (a), the derivation should be a dedicated appendix (target: 5–10 pages) covering: the FLRW-restricted ADM ansatz, the homogeneous/isotropic burden tensor components, the S_{eff} variation step-by-step, and the κ_B calibration.

Fix difficulty: Hard (requires executing the FLRW ADM variation — days to weeks of focused work) | **Blocks Phase C:** Yes — without this, Λ CDM recovery (Q-13) remains PARTIAL and cosmological predictions remain heuristic

Change-log cross-ref: Flagged

The v1.0→v2.0 change log (Ch 4.7, Change 1) correctly noted that v2.0 honestly downgrades the Friedmann equation to a Conjecture with the derivation path stated but not performed. This is the second of the two issues the change log did catch.

► [M6] §2–§3 interdependence tighter than acyclicity suggests

Severity: MATERIAL | *Reviewers:* R3 | *v2.0 location:* §2.3.5 (uses K_{ω} from §3.1) + §3.8.2 (uses stable field modes from §2.4) + §3.9 (uses sectors from §1.2/§2.1.5)

Problem. While the manuscript claims acyclicity (Thm 7.5.1), the mutual referencing between §2 (fields/particles) and §3 (geometry) is tight: §2.3.5 uses K_{ω} (from §3.1), §3.8.2 uses stable field modes (from §2.4), §3.9 uses sectors (from §1.2/§2.1.5). In the presented order (§2 before §3), this is acyclic. However, the logical structure treats geometry and microscopic dynamics as interdependent. The 'smooth field convergence' conjecture (Conj 3.8.2) even suggests the smooth manifold EMERGES from field mode density — a reversal of the standard hierarchy that requires both §2 and §3 to hold simultaneously. This is not a formal circularity, but the two layers are not as cleanly separable as the architecture suggests.

Fix. Acknowledge the interdependence explicitly. Add a remark to §3.0 or §7.5: 'The dependency graph is acyclic in the strict sense (no object is used before its ingredients exist), but §2 and §3 are co-emergent in a weaker sense: the smooth manifold (Conj 3.8.2) emerges from stable field mode density (§2.4), while the field admissibility condition (§2.3.1) uses K_ω (§3.1). The one-way references in the presented order are sufficient for acyclicity, but the physical picture is one of co-emergence, not strict hierarchy.' Consider reframing the smooth-field-convergence conjecture as 'co-emergent' rather than 'emergent from §2.'

Fix difficulty: Medium (1-paragraph remark + conceptual reframing) | Blocks Phase C: No — but the honest commentary must be present

Change-log cross-ref: Not flagged

The v1.0→v2.0 change log did not flag the §2–§3 interdependence. The change log was organized section-by-section and did not assess cross-section dependency density.

► [M7] T-1 status generous — Thm 1.1.2 is dimensional argument, not derivation

Severity: MATERIAL | Reviewers: R3 | v2.0 location: Thm 1.1.2 (finite propagation rate $c = \gamma \cdot \ell_0$) + T-1 (γ derivation from spectral gap)

Problem. Theorem 1.1.2 (finite propagation rate $c = \gamma \cdot \ell_0$) is labeled 'Conditional' with T-1 (derivation of γ) as the blocking dependency. However, the proof sketch is not a derivation from deeper principles; it is a dimensional argument ('each incremental expansion requires time at least $\epsilon = 1/\gamma$ '). The status '[Conditional]' is generous — this is closer to a heuristic scaling relation than a derived bound. A conditional theorem should have a proof sketch that would go through if the blocking target were closed; Thm 1.1.2 doesn't have that.

Fix. Downgrade Theorem 1.1.2 from 'Conditional' to 'Heuristic Scaling Relation' or 'Structural Postulate' until T-1 closes. Add a remark: 'The bound $c = \gamma \cdot \ell_0$ is a dimensional argument: each incremental Open Expansion requires time at least $\epsilon = 1/\gamma$ at the fundamental scale ℓ_0 . A rigorous derivation requires T-1 (the spectral-gap derivation of γ from F). Until T-1 closes, this is a structural postulate, not a conditional theorem.'

Fix difficulty: Quick win (status re-tag + 1-paragraph remark) | Blocks Phase C: No — but must be fixed for honest status labels

Change-log cross-ref: Not flagged

The v1.0→v2.0 change log (Ch 4.2) noted that v2.0 §1 introduces the finite rate $c = \gamma \cdot \ell_0$ and flags T-1 as the blocking dependency, but did not evaluate whether the proof sketch is strong enough to support 'Conditional' status.

► [M8] Open Extension Lindblad form T-2 dependency not propagated to downstream theorems

Severity: MATERIAL | Reviewers: R3 | v2.0 location: Eq 1.5.3b (Open Extension Lindblad form, T-2) + Thm 4.1.4 (Arrow of Time) + Thm 2.6.3 (decoherence)

Problem. The derived Lindblad form of the Open Extension flow (Eq 1.5.3b) is flagged as 'Theorem Target T-2.' Without T-2, the dissipative structure of R_t is not rigorously established. This dependency propagates to: the Arrow of Time proof (Thm 4.1.4, which uses Spohn's inequality on the Lindblad generator) and the decoherence mechanism (Thm 2.6.3, which uses the cross-sector burden from the Open Extension flow). But Thm 4.1.4 and Thm 2.6.3 do not currently carry [T-2] tags.

Fix. Add inline [T-2] tags to Thm 4.1.4 and Thm 2.6.3 (and any other theorem whose proof invokes the Lindblad form of Open Extension). Update Table 7.1.2 to show the T-2 dependency in the 'Depends On' column for these theorems. Add a remark to §1.5: 'The Lindblad form (Eq 1.5.3b) is T-2-conditional. Theorems whose proofs invoke this form — including Thm 4.1.4 (Arrow of Time) and Thm 2.6.3 (Stable

Record Separation) — inherit the T-2 dependency.'

Fix difficulty: Quick win (inline tags + 1-paragraph remark + table column update) | **Blocks Phase C:** No — but must be fixed for honest dependency tracking

Change-log cross-ref: Partially flagged

The v1.0→v2.0 change log (Ch 4.2, Change 5; Ch 4.5, Change 2) noted the T-2 dependency of R_t 's Lindblad form and the Arrow of Time proof's dependence on the correct generator assignment, but did not flag that the T-2 dependency is not propagated to Thm 4.1.4 and Thm 2.6.3 via inline tags.

Chapter 6 — Cosmetic Issues (2)

Two cosmetic issues. Plus one already-fixed note (C3) documenting progress on the A_{loc} selection language.

► [C1] LaTeX artifacts in equations

Severity: COSMETIC | Reviewers: R3 | v2.0 location: Eq 2.3.6 and possibly elsewhere

Problem. The manuscript still contains LaTeX artifacts (e.g., $\backslash\mathrm{nabla}\cdot\mathrm{f}\{\partial\}_{-}^2\mathrm{f}\{\Gamma\}\backslash\mathrm{f}\{\varphi\}$) in Eq 2.3.6). These are rendering errors where raw LaTeX source appears in the PDF instead of being compiled to mathematical notation.

Fix. Clean all LaTeX artifacts in the Phase C archival pass. Search for common artifact patterns: $\backslash\mathrm{f}\{\partial\}$, $\backslash\mathrm{f}\{\nabla\}$, $\backslash\mathrm{f}\{\Gamma\}$, $\backslash\mathrm{f}\{\varphi\}$, stray backslashes, uncompiled math mode delimiters. Re-render all equations with a consistent math typesetting pipeline.

Fix difficulty: Quick win (find-and-fix in archival pass) | Blocks Phase C: No — but must be fixed before external dissemination

Change-log cross-ref: Not flagged (out of scope for change log)

The v1.0→v2.0 change log was about structural changes, not typography. LaTeX artifacts are a production issue, not a structural one, so the change log did not survey for them.

► [C2] Thm 0.4.4 naming slightly misleading

Severity: COSMETIC | Reviewers: R1 | v2.0 location: Thm 0.4.4 (titled 'Complementarity / Principle of the Now')

Problem. Theorem 0.4.4 is titled 'Complementarity / Principle of the Now,' but the 'now' is not defined until §4.2 (after both space §3 and time §4.1 exist). The theorem text correctly limits its scope to 'governing mechanisms' (Lie + Jordan), so this is a naming issue rather than a structural problem — but a reader who sees 'Principle of the Now' in §0 will expect the now to be defined there, and will be confused when it is not.

Fix. Rename Thm 0.4.4 from 'Complementarity / Principle of the Now' to 'Complementarity of the Governing Mechanisms.' Drop 'Principle of the Now' from the theorem name; if desired, add a forward-reference remark: 'The operational "now" is defined at §4.2, after both the operational "where" (§3) and "when" (§4.1) are available.'

Fix difficulty: Quick win (rename + 1-line remark) | Blocks Phase C: No

Change-log cross-ref: Not flagged

The v1.0→v2.0 change log (Ch 4.1, Change 5) noted that v2.0 §0.4 introduces Theorem 0.4.4 (Complementarity) as the section's headline theorem, but did not flag the naming issue.

6.3 Already Fixed in v2.0 (Noted as Progress)

► [C3] A_{loc} selection language — FIXED in v2.0 (noted as progress)

Severity: ALREADY FIXED | Reviewers: R4 | v2.0 location: Def 3.1.1 ($A_{\text{loc}} = \{A \in A_{\text{phy}}^{\text{thin}} : \omega_{\text{kin}}(A \nmid A) > 0\}$)

What was fixed. v1.0's A_{loc} was 'a chosen class... different choices give different geometries' — a selection, not a derivation. v2.0's Def 3.1.1 makes A_{loc} fully determined: $A_{\text{loc}} = \{A \in A_{\text{phy}}^{\text{thin}} : \omega_{\text{kin}}(A^\dagger A) > 0\}$ — every element of $A_{\text{phy}}^{\text{thin}}$ with positive GNS norm, no selection involved. This directly answers the 'is the sector stipulated or inferred' question. NOT an open issue — listed as progress.

Action. None needed. Already fixed in v2.0. Ensure consistency throughout (no remaining 'chosen class' language).

Chapter 7 — Cross-Reference to v1.0→v2.0 Change Log

For each of the 15 open issues, the table below notes whether the v1.0→v2.0 change log (RCF_v1_to_v2_Change_Log.pdf, 48 pages) flagged it. The change log was descriptive (faithfully reported what v2.0 claims) but not evaluative (did not verify whether v2.0's self-descriptions are correct).

Status	Count	Issues
Flagged	2	M4 (κ_B calibrated), M5 (Friedmann not executed)
Partially flagged	4	F1 (T-2 listed but not surfaced as reader-facing issue), F5 (D=3 Ξ_C fix noted but three-channel assumption not flagged), M3 (arithmetic fix noted but convention fuzziness not), M8 (T-2 dependency noted but not propagated)
Not flagged	9	F2 (K_ω Jordan-built claim taken at face value), F3 (RP derivation gap not caught), F4 (T-4 / Z-envariance contradiction not connected), M1 (§1 forward-ref not checked against §2 statement), M2 (status inheritance not checked), M6 (cross-section density not assessed), M7 (proof sketch strength not evaluated), C1 (out of scope), C2 (naming issue not flagged)

Self-Assessment

The v1.0→v2.0 change log was descriptive (faithfully reported what v2.0 claims) but not evaluative (did not verify whether v2.0's self-descriptions are correct). Of 15 issues identified by the external reviews, the change log caught 2 (M4, M5), partially caught 4 (F1, F5, M3, M8), and missed 9 (F2, F3, F4, M1, M2, M6, M7, C1, C2). The most significant missed catches are F2 (K_ω not genuinely Jordan-built) and F4 (T-4 overclaim) — both are cases where v2.0 makes a structural claim and the change log reported the claim without checking whether it is earned. The lesson for future change logs: a structural rewrite's self-documentation (correction lists, preambles) is a starting point, not a substitute for independent verification.

7.1 Issue-by-Issue Cross-Reference

ID	Issue	Change-Log Status	Note (truncated)
F1	T-2 as single point of failure — not surfaced in abstract/in...	Partially flagged	The v1.0→v2.0 change log (Ch 6) listed T-2 as the existential open target and noted its blocking rol...
F2	K_ω is not genuinely built from the Jordan product	Not flagged	The v1.0→v2.0 change log (Ch 4.4, Change 10) noted that v2.0 §3 added Remark 3.1.3 'ties K_ω to the ...
F3	Reconciliation Principle not derived from Open Expansion	Not flagged	The v1.0→v2.0 change log (Ch 4.3) noted that RP was moved to §2.1 and that v2.0 frames it as 'the dy...

ID	Issue	Change-Log Status	Note (truncated)
F4	T-4 overclaim — CLOSED but Z-envariance depends on T-2	Not flagged	The v1.0→v2.0 change log (Ch 4.8, Change 4) noted that v2.0 says 'T-4 is CLOSED' and that Z-envarian...
F5	D=3 three-channel assumption is ad hoc	Partially flagged	The v1.0→v2.0 change log (Ch 4.4, Change 1) noted the D=3 demotion and the tautological Ξ_C fix, but...
M1	Mass-burden identity inconsistency across sections	Not flagged	The v1.0→v2.0 change log (Ch 4.3, Change 13) noted the $m \equiv B_0 \rightarrow m^2 \equiv B_0$ fix in §2.5.1, but did not f...
M2	Dark-energy cosmological predictions status too strong	Not flagged	The v1.0→v2.0 change log (Ch 4.7) noted the dark energy mechanism and the $(1+z)^{-6}$ suppression as ...
M3	Quarantine arithmetic convention not finalized	Partially flagged	The v1.0→v2.0 change log (Ch 4.8, Change 3) noted the arithmetic fix (13→11 PARTIAL), but did not fl...
M4	$\Pi_{\max}$, κ_B , and $C=8\pi$ presented as 'derived' in some summarie...	Flagged	The v1.0→v2.0 change log (Ch 4.5, Change 8; Ch 4.6, Change 3) correctly noted that v2.0 presents κ_B ...
M5	Friedmann derivation not executed	Flagged	The v1.0→v2.0 change log (Ch 4.7, Change 1) correctly noted that v2.0 honestly downgrades the Friedm...
M6	§2–§3 interdependence tighter than acyclicity suggests	Not flagged	The v1.0→v2.0 change log did not flag the §2–§3 interdependence. The change log was organized sectio...
M7	T-1 status generous — Thm 1.1.2 is dimensional argument, not...	Not flagged	The v1.0→v2.0 change log (Ch 4.2) noted that v2.0 §1 introduces the finite rate $c = \gamma \cdot \ell_0$ and flags T...
M8	Open Extension Lindblad form T-2 dependency not propagated t...	Partially flagged	The v1.0→v2.0 change log (Ch 4.2, Change 5; Ch 4.5, Change 2) noted the T-2 dependency of R_t 's Lind...
C1	LaTeX artifacts in equations	Not flagged (out of scope for change log)	The v1.0→v2.0 change log was about structural changes, not typography. LaTeX artifacts are a product...
C2	Thm 0.4.4 naming slightly misleading	Not flagged	The v1.0→v2.0 change log (Ch 4.1, Change 5) noted that v2.0 §0.4 introduces Theorem 0.4.4 (Complemen...

Chapter 8 — v2.1 Patch Execution Checklist

Ordered execution checklist for v2.1, sorted by severity (fatal first, then material, then cosmetic), then by fix difficulty within severity (quick wins first). Each item: issue ID, patch description, target section, estimated effort, and whether it blocks Phase C.

ID	Patch	Target Section	Effort	Blocks Phase C?
F4	Demote T-4 from CLOSED to STRENGTHENED (conditional on T-2) in Table 7.2.1; add 1-paragraph remark to §7.2.3	§7.2.3 + Table 7.2.1	Quick win	No
F1	Add T-2 dependency callout to abstract + introductions of §4, §5, §6; add inline [T-2] tags to every T-2-conditional theorem	Abstract + §4–§6 intros + theorem statements	Medium	Yes
F5	Expand Remark 3.6.5 into 1-page discussion of why the three-channel assumption is not derivable; consider further demotion to 'Structural Conjecture'	§3.6 Remark 3.6.5	Medium	No
F3	Either derive RP from Open Expansion (hard) or explicitly add RP to the primitive list in Guardrail 4 (§7.6.1) (easy)	Princ 2.1.1 + Guardrail 4 (§7.6.1)	Easy fallback / Hard derivation	Yes
F2	Build K_ω -analogue with $A \circ B$ in the formula (preferred) OR honestly downgrade Remark 3.1.3 to 'symmetry from state-Hermiticity, Jordan structure licenses interpretation only' (fallback)	Def 3.1.2 + Remark 3.1.3	Hard	Yes
M 1	Find-and-replace ' $m \equiv B_0$ ' → ' $m^2 \equiv B_0$ ' across all 8 sections; audit summary tables	All sections (especially §1 forward-ref)	Quick win	No
M 2	Add conditional tags to Thms 6.3.3–6.3.5 (CONDITIONAL on FLRW reduction); add 1-paragraph remark to §6.3	§6.3	Quick win	No
M 4	Audit summary tables and abstract for ' κ_B derived' / ' $\Pi_{\max}$ resolved' / ' $C = 8\pi$ derived'; replace with 'calibrated' / 'partially resolved'	Summary tables + abstract	Quick win	No
M 7	Downgrade Thm 1.1.2 from 'Conditional' to 'Heuristic Scaling Relation' or 'Structural Postulate'; add 1-paragraph remark	Thm 1.1.2	Quick win	No
M 8	Add inline [T-2] tags to Thm 4.1.4 and Thm 2.6.3; update 'Depends On' column in Table 7.1.2	Thm 4.1.4, Thm 2.6.3, Table 7.1.2	Quick win	No
M 3	Adopt three hard statuses (OPEN / STRUCTURAL / CONDITIONAL) for quarantine list; re-tag all 17 Q-items	§7.3 Remark 7.3.2 + Table 7.3.1	Medium	No

ID	Patch	Target Section	Effort	Blocks Phase C?
M 6	Add 1-paragraph remark to §3.0 or §7.5 acknowledging §2–§3 co-emergence; reframe Conj 3.8.2 as 'co-emergent'	§3.0 or §7.5	Medium	No
M 5	Execute the 4-step FLRW ADM variation (preferred) OR maintain 'Conjecture' status with explicit anti-drift remark	Conj 6.2.2 (or new appendix)	Hard	Yes
C1	Clean all LaTeX artifacts in archival pass (search for <code>\mathsf</code> , <code>\partial</code> , <code>\nabla</code> , stray backslashes)	All equations (especially Eq 2.3.6)	Quick win	No
C2	Rename Thm 0.4.4 from 'Complementarity / Principle of the Now' to 'Complementarity of the Governing Mechanisms'	Thm 0.4.4	Quick win	No

8.1 Minimum Viable v2.1

Of the 15 open issues, 9 are mandatory for v2.1 to honestly be called 'canonical,' and 6 can wait for v2.2 / Phase C.

Mandatory for v2.1 (9 patches)

Patches: F4, F1, M1, M2, M4, M7, M8, C1, C2.

These 9 patches are either quick wins (1-line edits, table re-tags, find-and-replace) or medium-effort honesty fixes that do not require new derivations. They bring v2.0's status labels into internal consistency and surface the T-2 dependency prominently. Without these, v2.0 cannot honestly be called 'canonical.'

Can Wait for v2.2 / Phase C (6 patches)

Patches: F2, F3, F5, M3, M5, M6.

These 6 patches require either hard derivations (F2 K_ω Jordan-built, F3 RP derivation, M5 Friedmann execution) or medium-effort conceptual expansions (F5 $D=3$ three-channel discussion, M3 quarantine re-tagging, M6 co-emergence reframing). They are real issues but do not block v2.1's claim to honest status labels. They should be prioritized for v2.2 / Phase C.

8.2 Recommended Execution Order

If the goal is to ship v2.1 as quickly as possible while maintaining honesty, execute in this order: (1) all quick wins first (F4, M1, M2, M4, M7, M8, C1, C2) — these are 1-line edits or table re-tags that bring v2.0's status labels into internal consistency; (2) then the medium-effort honesty fixes (F1 T-2 callout, F5 $D=3$ discussion, M3 quarantine re-tagging, M6 co-emergence remark) — these add the reader-facing honesty that the quick wins prepare the ground for; (3) defer the hard patches (F2 K_ω Jordan-built, F3 RP derivation, M5

Friedmann execution) to v2.2 / Phase C, with explicit anti-drift remarks so their status doesn't inflate in the meantime.

The single highest-leverage patch is **F1** (T-2 callout). It doesn't close T-2 — only a proof can do that — but it makes the framework's epistemic posture honest: every reader of v2.1 will know that the GR-level predictions are T-2-conditional, and can evaluate the framework accordingly. F1 + F4 + M1 + M2 + M4 + M7 + M8 + C1 + C2 (the 9 mandatory patches) take v2.0 from 'architecturally sound, status labels not fully earned' to 'architecturally sound, status labels honest.' That is the minimum bar for calling v2.1 canonical.